

ESERCIZIO

$$E = \{x \in \mathbb{R} : |x-3| \leq 3\} \setminus \{4\}$$

$$|x-3| \leq 3 \rightarrow \text{CASO 1} \begin{cases} x-3 \geq 0 \\ x-3 < 0 \end{cases}$$

CASO 1

$$x-3 \geq 0$$

$$x \geq 3$$

$$\downarrow$$

$$x-3 \leq 3$$

$$\downarrow$$

$$-3 \leq x-3 \leq 3$$

$$0 \leq x \leq 6$$

$$E = [0, 4) \cup (4, 6]$$

CASO 2

$$x-3 < 0$$

$$x < 3$$

$$\downarrow$$

$$-x+3 \leq 3$$

$$\downarrow$$

$$x-3 \geq -3$$



$$|x-a| \leq b \quad \text{con } b > 0$$

$$\downarrow$$

$$-b \leq x-a \leq b$$

ESERCIZIO

$$E = \{x \in \mathbb{R} : |x-3| < 2x\}$$

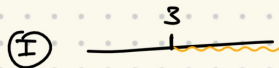
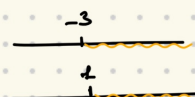
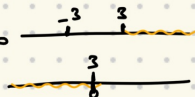
$$-2x < x-3 < 2x$$

$$\textcircled{1} \quad x-3 \geq 0 \rightarrow x \geq 3 \quad \text{segno rimane invariato}$$

$$\textcircled{2} \quad x-3 < 0 \rightarrow x < 3 \rightarrow -x+3 < 2x$$

$$\textcircled{1} \quad x-3 < 2x \rightarrow -3 < x$$

$$\textcircled{2} \quad -x+3 < 2x \rightarrow 3 < 3x \rightarrow 1 < x$$



$$(1, 3) \cup [3, +\infty)$$

PERCHÉ NON
PRENDIAMO
 $x > -3$

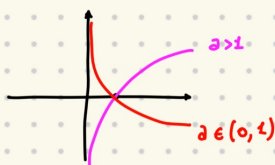
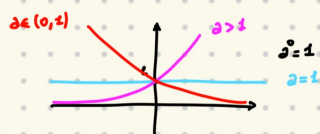
ESPOENZIALE

$$x = \sqrt[q]{a}, \quad a^x := \sqrt[q]{a^x}$$

FUNZIONE INVERSA
LOGARITMO

$$\log_a x$$

$$\log_a b = c$$



PROPRIETÀ DELL'ESPOENZIALE

$$I. \quad a^0 = 1$$

$$II. \quad a^x a^y = a^{x+y}$$

$$III. \quad (a^x)^y = a^{xy}$$

$$IV. \quad a > 1 \Rightarrow \text{FUNZIONE CRESCENTE}$$

$$a \in (0, 1) \Rightarrow \text{FUNZIONE DECRESCENTE}$$

PROPRIETÀ DEL LOGARITMO

$$I. \quad \log_a 1 = 0$$

$$II. \quad \log_a (xy) = \log_a x + \log_a y$$

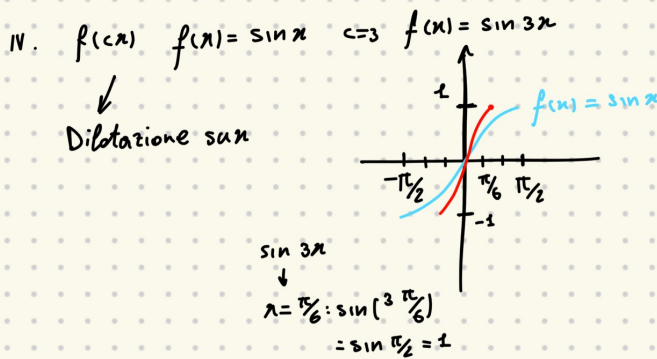
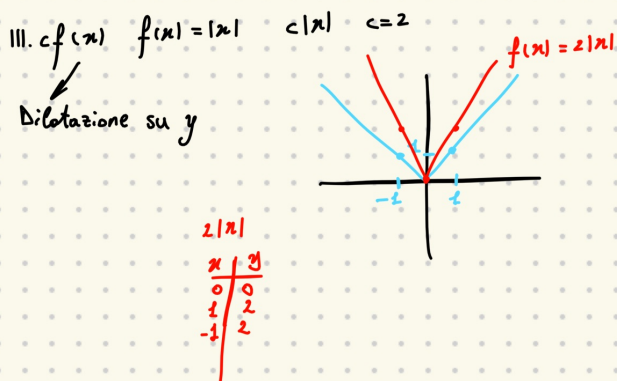
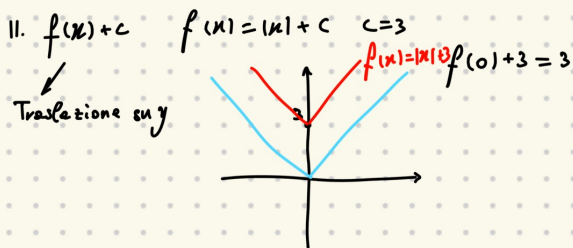
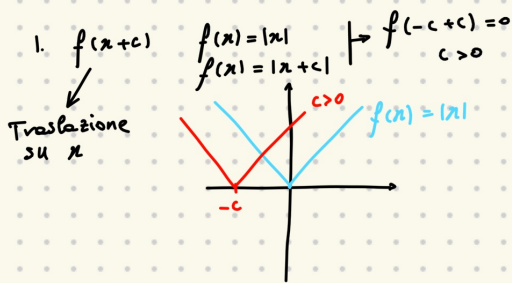
$$III. \quad \log_a x^k = k \log_a x$$

$$IV. \quad a > 1 = \text{FUNZIONE CRESCENTE}$$

$$a \in (0, 1) = \text{FUNZIONE DECRESCENTE}$$

$$V. \quad \text{Dominio } \log_a = (0, +\infty)$$

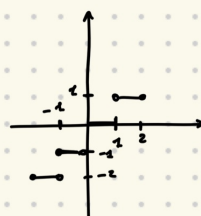
OPERAZIONI E GRAFICI DI FUNZIONI



$$f(x) = \frac{x+1}{x+2} \rightarrow f(x) = \frac{a_0 + a_1x + \dots + a_nx^n}{b_0 + b_1x + \dots + b_mx^m} \rightarrow \text{FUNZIONI RAZIONALI}$$

FUNZIONE PARTE INTERA $[x] = Lx$

$[x] = \mathbb{R} \ni x \mapsto \text{L'UNICO INTERO } z \in \mathbb{Z} : z \leq x < z+1$

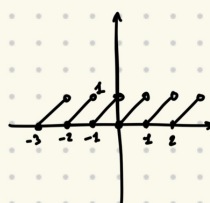


FUNZIONE MANTISSA $\{x\}$

$$\{x\} := x - [x]$$

T MANTISSA = 1

$$\{x\} \in [0, 1)$$

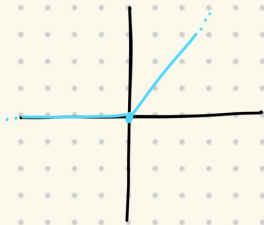


FUNZIONE PERIODICA

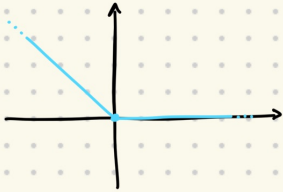
$f(x)$ è periodica se $\exists T > 0$
 $f(x) = f(x) + T$
 $\forall x$

FUNZIONE PARTE POSITIVA

$$P(x) = \max\{0, x\}$$

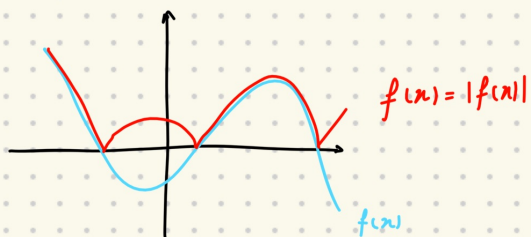


FUNZIONE PARTE NEGATIVA $N(x) = \max\{0, -x\}$

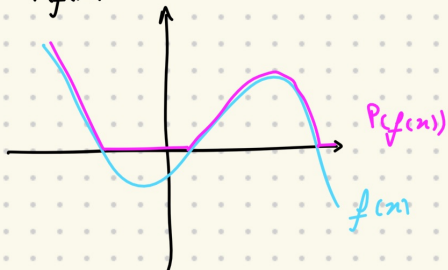


Quindi $f(x) = |x| = P(x) + N(x)$

$|f(x)|$



$P(f(x))$



$N(f(x))$

